The examination is being carried out on the **following application documents**

**Description, Pages**

| | | |
|---|---|---|
| 1, 3-72 | as published | |
| 2, 2a | filed in electronic form on | 07-08-2014 |

**Sequence listings, SEQ ID NO**

| | | |
|---|---|---|
| 1, 2 | as published | |

**Claims, Numbers**

| | | |
|---|---|---|
| 1-8 | filed in electronic form on | 07-08-2014 |

**Drawings, Sheets**

| | | |
|---|---|---|
| 1-15 | as published | |

## 1 CITED DOCUMENTS

The following **documents** are referred to in this communication; the numbering will be adhered to in the rest of the procedure:

D2 WO 2008/011193 A2 (GENERA DOO [HR]; VUKICEVIC SLOBODAN [HR]; GRGUREVIC LOVORKA [HR]; MACE) 24 January 2008 (2008-01-24)

D3 JINGYING WANG ET AL: "Limiting Cardiac Ischemic Injury by Augmenting the MIF-AMPK Signaling Cascade with a Novel Class of MIF Receptor Agonists",
CIRCULATION,
vol. 21, Suppl. S, 23 November 2010 (2010-11-23), page A15835, XP055097063,

D4 COLEMAN ARLIXER M ET AL: "Cooperative regulation of non-small cell lung carcinoma angiogenic potential by macrophage migration inhibitory factor and its homolog, D-dopachrome

tautomerase",
THE JOURNAL OF IMMUNOLOGY, THE AMERICAN ASSOCIATION OF IMMUNOLOGISTS, US,
vol. 181, no. 4, 1 August 2008 (2008-08-01), pages 2330-2337, XP002664500,
ISSN: 0022-1767

D5 MERK MELANIE; ZIEROW SWEN; LENG LIN; SCHULTE WIBKE; LESUR OLIVIER; BUCALARICHARD: "First functional characterization of a new MIF family member as a mediator of sepsis",
INFLAMMATION RESEARCH,
vol. 59, no. Suppl. 1, 9 March 2010 (2010-03-09), - 13 March 2010 (2010-03-13), pages s120-s121, XP008166736,

D7 DAN XIN ET AL: "The MIF Homologue D-Dopachrome Tautomerase Promotes COX-2 Expression through beta-Catenin-Dependent and -Independent Mechanisms",
MOLECULAR CANCER RESEARCH, AMERICAN ASSOCIATION FOR CANCER RESEARCH, US,
vol. 8, no. 12, 1 November 2010 (2010-11-01), pages 1601-1609, XP002664499,
ISSN: 1541-7786, DOI: 10.1158/1541-7786.MCR-10-0101
[retrieved on 2010-11-11]

The following document is cited by the Applicant.

D9 Dake Qi ET AL: "The vestigial enzyme D-dopachrome tautomerase protects the heart against ischemic injury",
Journal of Clinical Investigation,
vol. 124, no. 8, 1 July 2014 (2014-07-01), pages 3540-3550, XP055138130,
ISSN: 0021-9738, DOI: 10.1172/JCI73061

2 AMENDMENTS (Art. 123(2) EPC)

2.1 The amendments introduced are in accordance with Art. 123(2) EPC.

## 3 NOVELTY (Art. 54 EPC)

3.1 Claim 1 is directed to a composition comprising a D-DT polypeptide for treating a subject, a tissue or an organ. The embodiment wherein a tissue or an organ is treated is interpreted as opposed to treating a subject as an in vitro treatment. This is also clear from the passage on p. 43, lines 17-19. In this case, the subject-matter is not entitled to the second medical use formulation of Art. 54(4) EPC and the claim is read as merely directed to the composition disregarding the intended use. For this reason, any document disclosing the D-DT protein, such as any of D4, D5 and D7, is novelty destroying for this embodiment of claim 1. The embodiment should be claimed separately as a method of treatment.

3.2 The present application does not meet the requirements of Article 52 (1) EPC because the subject-matter of claim 1 is not new in the sense of Article 54 (1) and (2) EPC.

## 4 INVENTIVE STEP (Art. 56 EPC)

4.1 Document D2 is considered to represent the most relevant state of the art and discloses a method of treating ischemia/reperfusion damage to a kidney in an individual comprising: administering to the individual one or more antibody molecules specific for one or more BMP-I isoforms (cl. 28), wherein the one or more antibody molecules to one or more BMP-I isoforms is administered systemically to the individual prior to an ischemia/reperfusion event (cl. 29). The subject-matter of claim 1 differs in that the compound is one of D-DT polypeptide or a recombinant D-DT polypeptide. No the technical effect could be identified from this difference.

The problem to be solved by the present invention may therefore be regarded as providing is the provision of an alternative compound for treating or preventing ischemia-reperfusion. The proposed solution is one of D-DT polypeptide or a recombinant D-DT polypeptide.

This solution can be considered as involving an inventive step for the following reasons: D3 discloses the cardioprotective action of an agonist of MIF during ischemia and reperfusion and suggests that MIF receptor agonism by a small

molecule may provide protection against cardiac ischemic injury. Although MIF is a direct homologue of D-DT, there is no direct hint in the prior art that would link D-DT activity to ischemia protection instead of MIF.

Therefore, the proposed solution cannot be considered obvious and the claim is inventive.

4.2 In view of the lack of support (see 6.3), no inventive step can be acknowledged to the subject-matter of claim 2 or of its dependent claims 3-8.

## 5 UNPATENTABLE SUBJECT-MATTER (Art. 53 EPC)

5.1 Claim 8 refers to a method step comprising treating a subject. This falls under the exclusions from patentability contemplated in Art. 53(c) EPC. Hence, the claim should be deleted.

## 6 SUPPORT and DISCLOSURE (Art. 84, 83 EPC)

6.1 Claim 1 refers to a D-DT activator. This term does not fulfill the requirements of Articles 83 and 84 EPC. The term encompasses any agent having an effect anywhere upstream of the protein or gene. The claimed agent could modulate expression, be it directly or indirectly, or effect the function be it by direct interaction or indirectly. It would represent an undue burden to isolate and characterize all potential activators, without any effective pointer to their identity (Guidelines F-III,1) or to test every known compound and every conceivable future compound for the desired effect to see if it falls within the scope of the claims (Guidelines F-III, 9).

6.2 Claim 1 refers to an active D-DT polypeptide fragment. Neither the application or the prior art however discloses any example of an active D-DT fragment nor it provides sufficient indications about how to isolate such a fragment. Therefore, this embodiment is not considered sufficiently disclosed (Art. 83 EPC).

6.3 The application does not disclose a correlation between the endogenous expression or activity level of D-DT and the presence of an ischemia-reperfusion injury. Therefore claim 2 is not supported by the description (Art. 84 EPC). The post-published document D9 shows that a depletion of D-DT increases susceptibility to a stressful response to ischemia-reperfusion but not that low levels of D-DT can be used as a marker to detect an ongoing disorder. In other words, the susceptibility to a disease does not imply the presence of the disease itself. Therefore the disclosure of D9 cannot be used in support of the protection sought by claim 2.

## 7 CLARITY (Art. 84 EPC)

7.1 The expression in claim 1 "the composition for use comprising: administering..." is not clear (Art. 84 EPC) because it seems to define the composition by a method step. The claim should rather read "the composition comprising a D-DT polypeptide..." or even better "A composition comprising a D-DT polypeptide for use in a method of treating...".

## 8 CONCLUSIONS

8.1 The applicant is requested to file new claims which take account of the above comments.

8.2 When filing amended claims the applicant should at the same time bring the description into conformity with the amended claims. Care should be taken during revision, especially of the introductory portion and any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

8.3 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.